

SENTHIL THANNEERMALAI

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EDUCATION

Rice University, Houston, TX

May 2025

Master of Computer Science GPA: 3.96/4.0

- Relevant Coursework: Deep Learning for Vision and Language, Software Engineering Methodology, Analysis of Algorithms

University of Mumbai, Mumbai, India

Jun 2021

Bachelor of Engineering in Computer Engineering CGPA: 9.42/10

- Member of Computer Society of India, Winner of Best Web Development project for Movie Recommendation System

EXPERIENCE

Bloomberg Industry Group, Virginia

Jul 2025– Present

Software Engineer (Cursor, VueJS, Ruby on Rails, AWS, Terraform, Puppet)

- Enhanced the Vue.js frontend to streamline content creation and real-time editing workflows for Bloomberg Publishing, improving editor efficiency by over 20% through AI-powered autosuggestions and smart content formatting.
- Optimized Ruby on Rails backend for robust content management and efficient delivery of over 1 million monthly news articles.
- Managed AWS infrastructure for the publishing team using Puppet and Terraform, achieving 99.99% uptime and accelerating infrastructure deployments by 40%.
- Spearheaded the development of all Grafana Dashboards for the Publishing team, resulting in a 40% reduction in average error-finding time and significantly enhanced system tracing capabilities.

Maison and Tavola, Houston

May 2024 – May 2025

Web Optimization Intern (GraphQL, Flutter, Figma, Angular JS, Shopify)

- Implemented tools to automate tasks related to e-commerce website maintenance, boosting workflow efficiency by 80%.
- Improved website loading speed by 30% and enhanced accessibility by optimizing LCP, CLS, and INP metrics.
- Developed a robust, cross-platform iOS mobile application leveraging the Shopify API, featuring intelligent data fetching, responsive UI design, and secure account deletion with OAuth 2.0 authentication, set to debut on the App Store this year

Reliance Jio, Mumbai, India

Aug 2021 – Jul 2023

Software Development Engineer (RESTful APIs, React, Jenkins, Jest, JIRA, Redux)

- Architected and built a scalable drag-and-drop frontend using React and TypeScript for a consumer-based chatbot creation website, reducing chatbot creation time by 60% and eliminating the need for coding expertise through a test-driven approach
- Maintained a 99.9% uptime rate by rigorously testing, debugging, and resolving performance bottlenecks. Utilized Redis for efficient caching and session management, along with automated test cases to ensure uninterrupted platform functionality
- Led daily Agile scrum meetings, participated in code reviews, retrospect and collaborated and resolved technical stops for the team in both frontend and backend which led to the successful delivery of 3 major project milestones ahead of schedule
- Built unit tests and automated deployment pipelines using Jenkins, Docker, and AWS cloud-based CI/CD tools, reducing deployment time by 40% while optimizing application scaling and infrastructure costs

Bhabha Atomic Research Center, Supercomputing Facility, Mumbai, India

Dec 2019 – Jan 2020

Full-Stack Intern (Flask, SQL, React.js)

- Engineered a Management Information Base (MIB) Browser from the ground up, capable of parsing complex MIB files into an interactive interface and integrated with a custom Intrusion Detection System (IDS) for use across over 10 critical applications.

PROJECTS

Rapid Multimodal Query Resolution Using Nearest Neighbor Search (Docker, Flask, React.js) [\[view\]](#)

Jan 2024 – May 2024

- Engineered an advanced visual content search system integrating images, text, and audio, achieving sub-second response times using ANNOY, USearch, and FAISS algorithms, reducing search latency by 70%.
- Integrated SigLIP model to extract features from 1 million images in the SBU Captions dataset, improving search by 25%.
- Deployed multilingual search capabilities, supporting queries in 5 languages, increasing global accessibility by 30%

Automatic Report Generation Using Chest X-Ray Images (Python, Flask) [\[view\]](#)

Aug 2023 – Dec 2023

- Led a team of 5 in the development of an advanced neural network model with a co-attention mechanism and integrating ChexNet, GloVe embeddings and BioBERT, showcasing a 30% accuracy enhancement in medical report generation.

Attention Span Detection of Online Lecture (Python, Flask, React.js, Kafka) [\[view\]](#)

Aug 2020 – Jul 2021

- Designed an attention scoring mechanism that calculates attention score with an accuracy of 85%, leveraging Kafka for real-time processing of visual cues and background noise
- Published this project as an [IEEE paper](#), selected in top 10 among 100+ papers and presented it in [ICACC](#)

SKILLS

- Languages:** Python, C/C++, Java, HTML, CSS, Javascript, SQL
- Libraries & Tools:** GraphQL, Flutter, Flask, Shopify, Node, Git, Ruby on Rails, Terraform, React.js, VueJS, Postman, Redux, JIRA, Kafka, Jest, Express, Linux, Figma, REST APIs, PyTorch, OpenCV, NLTK, Matplotlib, Selenium
- Cloud:** AWS, GCP
- Deployment & CI/CD:** Docker, Kubernetes, Jenkins
- Databases:** MySQL, SQLite, PostgreSQL, Firebase, Firestore, Redis, NoSQL